

Pearson Edexcel International GCSE

# Mathematics A

## Modular Unit 1

### November 2023 Reconstructed Past Paper

#### DETAILED SOLUTIONS

Original question followed by a full worked solution

31 adaptive question sections    87 marks

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Selected from Linear Higher Papers 1H and 2H

## Dr Eslam Ahmed

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#### L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

#### Assistant Lecturer, Faculty of Engineering, Cairo University

B.Sc. Electrical Engineering (2017), Distinction with Honours

M.Sc. Mathematics (2023)

PhD Candidate

[eliteigcse.com](https://eliteigcse.com) • WhatsApp +20 112 000 9622

## About this reconstructed paper

Each question section appears in the same order as the Student book: the complete source demand is followed by its demand-local worked solution and final answer.

**Ownership rule:** Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 1; the selection contains 87 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

## Contents

| No. | Source      | Parts    | Marks | Page |
|-----|-------------|----------|-------|------|
| 1   | 4MA1-1H Q1  | (a)      | 1     | 3    |
| 2   | 4MA1-1H Q1  | (b)      | 1     | 5    |
| 3   | 4MA1-1H Q1  | (c)      | 1     | 7    |
| 4   | 4MA1-1H Q2  | (a)      | 2     | 9    |
| 5   | 4MA1-1H Q2  | (b)      | 4     | 11   |
| 6   | 4MA1-1H Q4  | (a)      | 2     | 13   |
| 7   | 4MA1-1H Q4  | (b)      | 2     | 15   |
| 8   | 4MA1-1H Q5  |          | 3     | 17   |
| 9   | 4MA1-1H Q10 |          | 5     | 19   |
| 10  | 4MA1-1H Q11 |          | 3     | 21   |
| 11  | 4MA1-1H Q12 | (a), (b) | 4     | 23   |
| 12  | 4MA1-1H Q12 | (c)      | 1     | 26   |
| 13  | 4MA1-1H Q17 |          | 3     | 28   |
| 14  | 4MA1-1H Q20 |          | 4     | 30   |
| 15  | 4MA1-2H Q2  | whole    | 4     | 32   |
| 16  | 4MA1-2H Q4  | (a)      | 3     | 34   |
| 17  | 4MA1-2H Q4  | (b)      | 1     | 36   |
| 18  | 4MA1-2H Q4  | (c)      | 2     | 38   |
| 19  | 4MA1-2H Q4  | (d)      | 2     | 40   |
| 20  | 4MA1-2H Q6  | whole    | 3     | 42   |
| 21  | 4MA1-2H Q9  | whole    | 2     | 44   |
| 22  | 4MA1-2H Q10 | whole    | 5     | 46   |
| 23  | 4MA1-2H Q11 | (a), (b) | 5     | 48   |
| 24  | 4MA1-2H Q12 | (a)      | 2     | 51   |
| 25  | 4MA1-2H Q12 | (b)      | 2     | 53   |
| 26  | 4MA1-2H Q15 | (a)      | 2     | 55   |
| 27  | 4MA1-2H Q15 | (b)      | 3     | 57   |
| 28  | 4MA1-2H Q16 | whole    | 3     | 59   |
| 29  | 4MA1-2H Q17 | whole    | 3     | 61   |
| 30  | 4MA1-2H Q20 | whole    | 4     | 63   |
| 31  | 4MA1-2H Q24 | whole    | 5     | 65   |

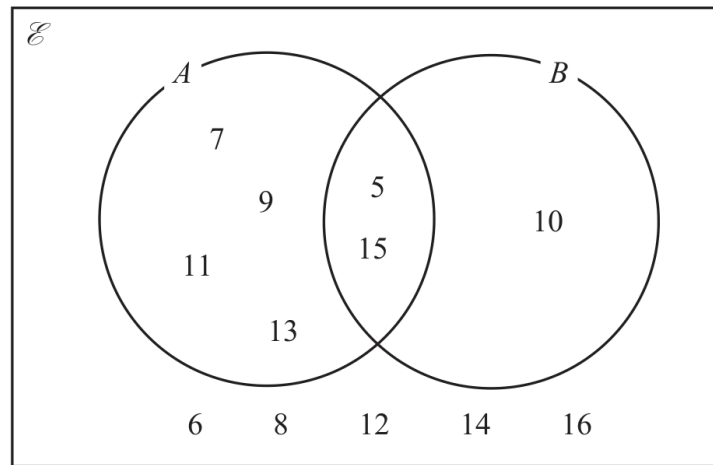
**Question 01**

4MA1-1H Q1 (a)

ORIGINAL QUESTION UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(a)  $A$

(1)

**Worked solution 01**

Read members of A

MODULAR UNIT 1

1 marks

(a) – Read members of A  
Read the diagram

$$A = 7, 9, 11, 13, 5, 15.$$

**Final answer**

$$A = \{5, 7, 9, 11, 13, 15\}$$

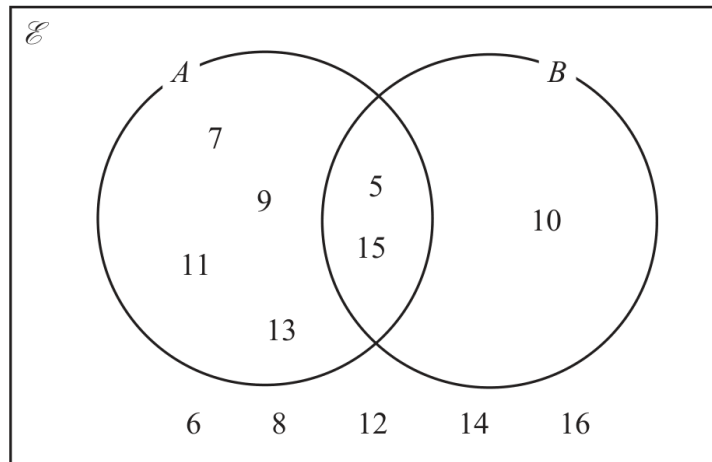
**Question 02**

4MA1-1H Q1 (b)

ORIGINAL QUESTION UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(b)  $A \cap B$

.....  
(1)

**Worked solution 02**

Read the intersection

MODULAR UNIT 1

1 marks

**(b) – Read the intersection**

Read the overlap

$$A \cap B = 5, 15.$$

**Final answer**

$$A \cap B = \{5, 15\}$$

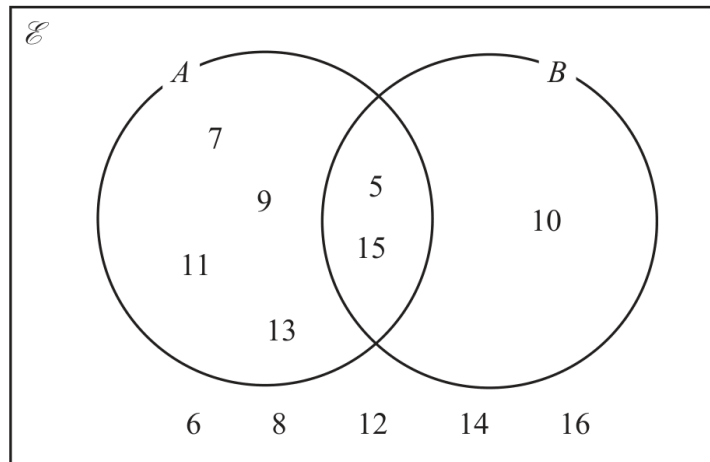
**Question 03**

4MA1-1H Q1 (c)

ORIGINAL QUESTION UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(c)  $(A \cup B)'$

.....  
(1)



**Worked solution 03**

Read the complement of the union

MODULAR UNIT 1

1 marks

**(c) – Read the complement of the union**

Read outside both circles

$$(A \cup B)' = 6, 8, 12, 14, 16.$$

**Final answer**

$$(A \cup B)' = \{6, 8, 12, 14, 16\}$$

**Question 04**

4MA1-1H Q2 (a)

ORIGINAL QUESTION UNIT 1

2 marks

2 (a) Factorise fully  $12pq - 18p$ .....  
(2)

**Worked solution 04**

Factorise fully

MODULAR UNIT 1

2 marks

**(a) – Factorise fully**

Take out the common factor

$$12pq - 18p = 6p(2q - 3).$$

**Final answer**

$$6p(2q - 3)$$

**Question 05**

4MA1-1H Q2 (b)

ORIGINAL QUESTION UNIT 1

4 marks

There are 56 metal bars in a box.  
Each metal bar is gold or silver or zinc.

$y$  metal bars are gold.

$(3y + 7)$  metal bars are silver.

$(2y - 5)$  metal bars are zinc.

- (b) Work out the number of metal bars that are zinc.  
Show clear algebraic working.

.....  
(4)

**Worked solution 05**

Find the number of zinc bars

MODULAR UNIT 1

4 marks

**(b) – Find the number of zinc bars**

Use the total

$$y + (3y + 7) + (2y - 5) = 56 \Rightarrow 6y + 2 = 56.$$

Solve for y

$$6y = 54 \Rightarrow y = 9.$$

Find zinc

$$2y - 5 = 2(9) - 5 = 13.$$

**Final answer**

13 zinc bars

**Question 06**

4MA1-1H Q4 (a)

ORIGINAL QUESTION UNIT 1

2 marks

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

|                    |      |        |       |
|--------------------|------|--------|-------|
| <b>Colour</b>      | blue | orange | white |
| <b>Probability</b> | 0.58 | $2x$   | $x$   |

- (a) Work out the value of  $x$

$x = \dots\dots\dots$   
(2)

**Worked solution 06**

Complete the spinner probabilities

MODULAR UNIT 1

2 marks

**(a) – Complete the spinner probabilities**

Use total probability

$$0.58 + 2x + x = 1 \Rightarrow 3x = 0.42.$$

Solve

$$x = 0.14.$$

**Final answer**

$$x = 0.14$$

**Question 07**

4MA1-1H Q4 (b)

ORIGINAL QUESTION UNIT 1

2 marks

- 4 A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

| Colour      | blue | orange | white |
|-------------|------|--------|-------|
| Probability | 0.58 | $2x$   | $x$   |

The spinner is spun 250 times.

- (b) Work out an estimate for the number of times the spinner will land on blue.

.....  
(2)



**Worked solution 07**

Expected blue outcomes

MODULAR UNIT 1

2 marks

**(b) – Expected blue outcomes**

Scale the blue probability

$$250(0.58) = 145.$$

**Final answer**

145

**Question 08**

4MA1-1H Q5 demand

ORIGINAL QUESTION UNIT 1

3 marks

- 5 The diagram shows a shaded shape made from three identical semicircles, *X*, *Y* and *Z*

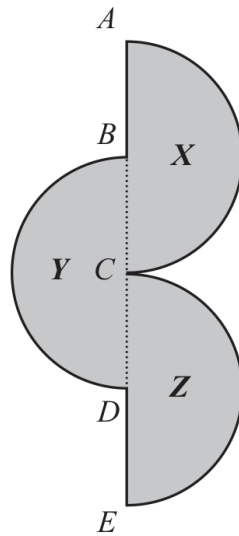


Diagram **NOT**  
accurately drawn

*ABCDE* is a straight line.

*AC* is the diameter of semicircle *X* and *B* is the centre of semicircle *X*

*BD* is the diameter of semicircle *Y* and *C* is the centre of semicircle *Y*

*CE* is the diameter of semicircle *Z* and *D* is the centre of semicircle *Z*

$$AC = BD = CE = 20 \text{ cm}$$

Work out the perimeter of the shaded shape.

Give your answer correct to the nearest whole number.

..... cm

**Worked solution 08**

Perimeter of three semicircles

MODULAR UNIT 1

3 marks

**Demand – Perimeter of three semicircles**

Add curved parts

$$3 \left( \frac{1}{2} \pi \times 20 \right) = 30\pi.$$

Add exposed diameters

$$AB + DE = 10 + 10 = 20; \quad P = 30\pi + 20 \approx 114.25.$$

**Final answer**

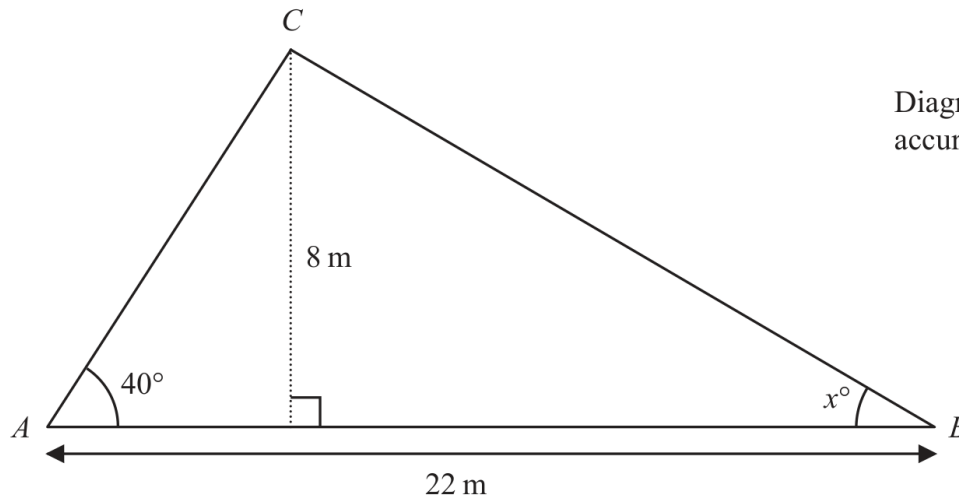
114 cm

**Question 09**

4MA1-1H Q10 demand

ORIGINAL QUESTION UNIT 1

5 marks

10 Here is triangle  $ABC$ Diagram **NOT**  
accurately drawn

Work out the value of  $x$   
Give your answer correct to one decimal place.  
Show your working clearly.

 $x = \dots\dots\dots$

**Worked solution 09**

Angle in a triangle from a perpendicular height

MODULAR UNIT 1

5 marks

**Demand – Angle in a triangle from a perpendicular height**

Find the left base segment

$$AD = \frac{8}{\tan 40^\circ} = 9.534\dots$$

Find the right base segment

$$DB = 22 - 9.534\dots = 12.466\dots$$

Use the right triangle at B

$$\tan x = \frac{8}{12.466\dots}$$

Evaluate

$$x = 32.690\dots^\circ \approx 32.7^\circ$$

**Final answer** $32.7^\circ$

**Question 10**

4MA1-1H Q11 demand

ORIGINAL QUESTION UNIT 1

3 marks

- 11 Express  $\frac{3}{4} + \frac{5-x}{6x}$  as a single fraction in its simplest terms.

.....

**Worked solution 10**

Single algebraic fraction

MODULAR UNIT 1

3 marks

**Demand – Single algebraic fraction**

Use a common denominator

$$\frac{3}{4} + \frac{5-x}{6x} = \frac{9x + 2(5-x)}{12x}.$$

Simplify

$$\frac{9x + 10 - 2x}{12x} = \frac{7x + 10}{12x}.$$

**Final answer** $(7x + 10)/(12x)$

**Question 11**

4MA1-1H Q12 (a), (b)

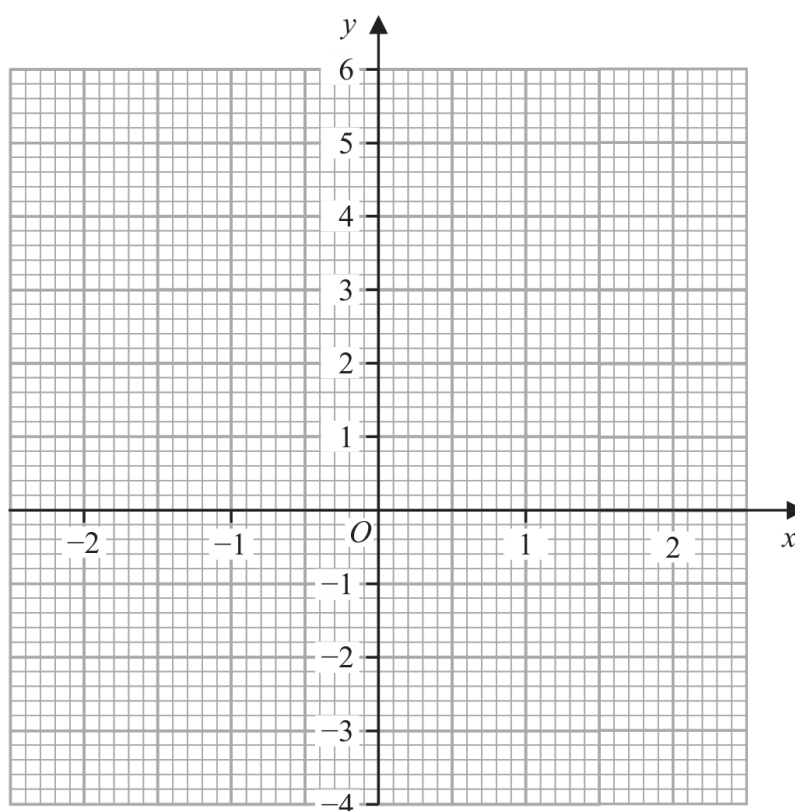
ORIGINAL QUESTION UNIT 1

4 marks

12 (a) Complete the table of values for  $y = x^3 - 3x + 1$ 

|     |    |    |   |    |   |
|-----|----|----|---|----|---|
| $x$ | -2 | -1 | 0 | 1  | 2 |
| $y$ |    |    |   | -1 |   |

(2)

(b) On the grid, draw the graph of  $y = x^3 - 3x + 1$  for values of  $x$  from -2 to 2

(2)



**Worked solution 11**

Complete the cubic table / Draw the cubic

MODULAR UNIT 1

4 marks

**(a) – Complete the cubic table**

Substitute each x

$$y = x^3 - 3x + 1 \Rightarrow y(-2, -1, 0, 1, 2) = (-1, 3, 1, -1, 3).$$

**Final answer**

y values: -1, 3, 1, -1, 3

**(b) – Draw the cubic**

Plot the table points

$$(-2, -1), (-1, 3), (0, 1), (1, -1), (2, 3).$$

Join smoothly

*Draw one smooth cubic curve through all five points for  $-2 \leq x \leq 2$ .*

**Final answer**Smooth cubic through  $(-2, -1), (-1, 3), (0, 1), (1, -1), (2, 3)$

**Completed diagram 11**

Complete the cubic table / Draw the cubic

MODULAR UNIT 1

4 marks

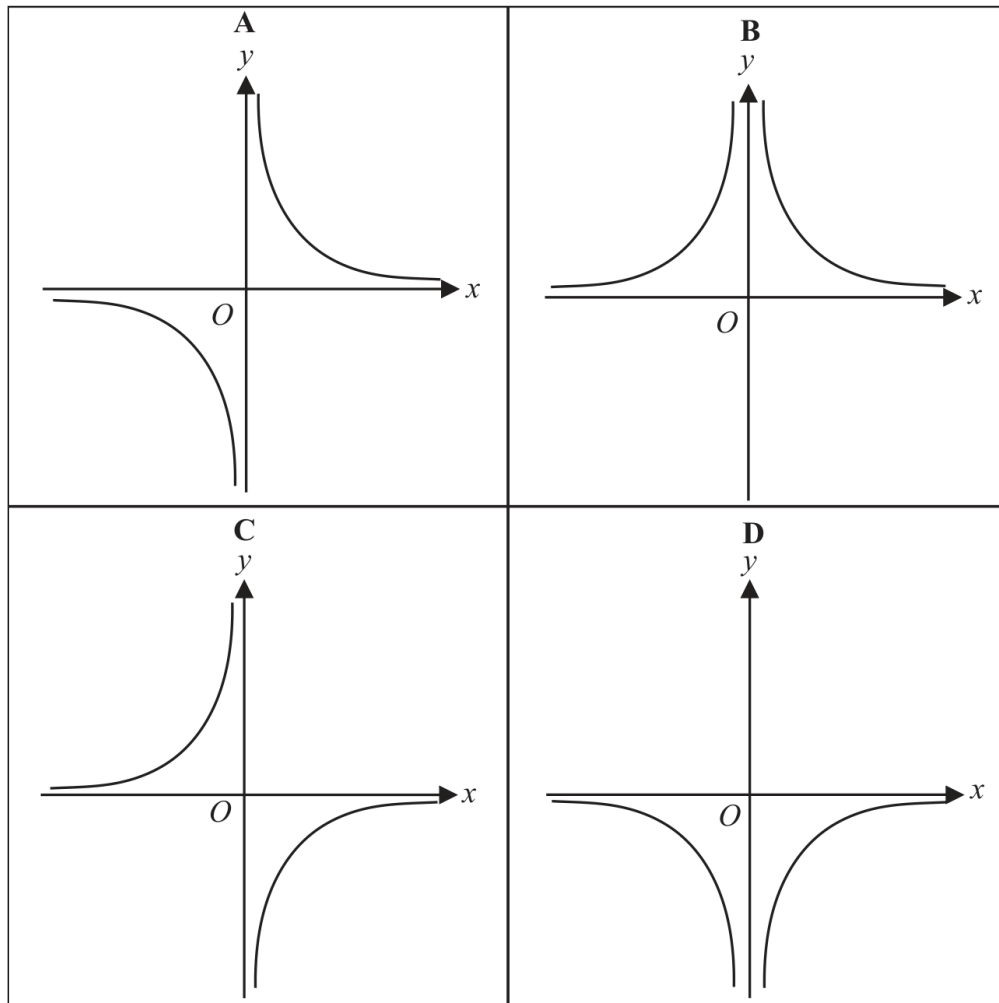
**Question 12**

4MA1-1H Q12 (c)

ORIGINAL QUESTION UNIT 1

1 marks

Here are four graphs.



(c) Write down the letter of the graph that could have the equation  $y = -\frac{1}{x^2}$

.....  
(1)

**Worked solution 12**Recognise  $y = -1/x^2$ 

MODULAR UNIT 1

1 marks

**(c) – Recognise  $y = -1/x^2$** 

Use the sign and asymptotes

 $y = -1/x^2 < 0$  for  $x \neq 0$ , with branches in quadrants III and IV; this is graph D.**Final answer**

D

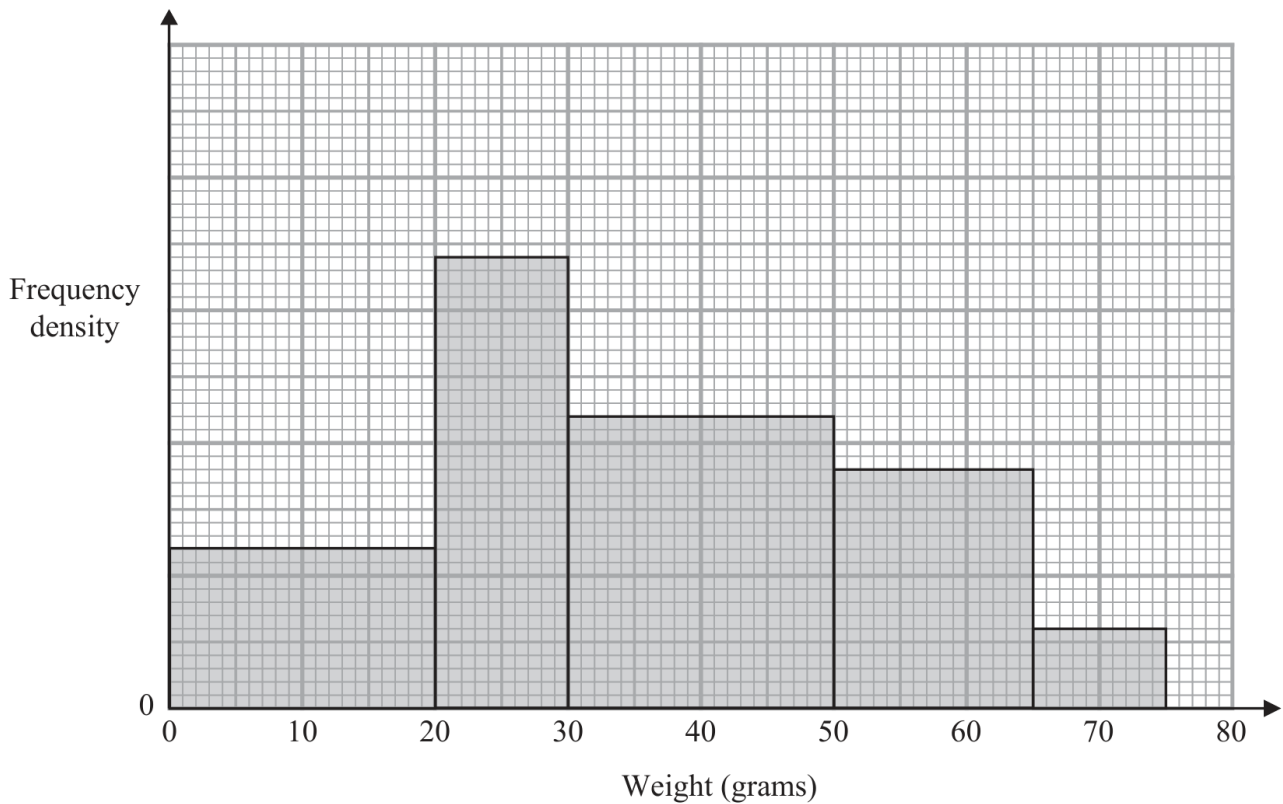
**Question 13**

4MA1-1H Q17 demand

ORIGINAL QUESTION UNIT 1

3 marks

17 The histogram gives information about the weights, in grams, of some oranges in a box.



24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

.....

**Worked solution 13**

Histogram area estimate

MODULAR UNIT 1

3 marks

**Demand – Histogram area estimate**

Calibrate the density scale

 $24/20 = 1.2$ ; the first bar therefore sets the small – square scale.

Add the two areas

$$35-50 : 15(2.2) = 33; \quad 50-55 : 5(1.8) = 9.$$

Total

$$33 + 9 = 42.$$

**Final answer**

42 oranges

**Question 14**

4MA1-1H Q20 demand

ORIGINAL QUESTION UNIT 1

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.  
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is  $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

**Worked solution 14**

Probability of mean 15 cents

MODULAR UNIT 1

4 marks

**Demand – Probability of mean 15 cents**

Find each coin probability

$$P(10) = \sqrt{49/121} = 7/11, \quad P(20) = 4/11.$$

Use the two orders

$$P(10, 20 \text{ or } 20, 10) = 2 \times \frac{7}{11} \times \frac{4}{11} = \frac{56}{121}.$$

**Final answer**

56/121



**Question 15**

4MA1-2H Q2 whole

ORIGINAL QUESTION UNIT 1

4 marks

- 2 The diagram shows triangle  $ABC$  and triangle  $ECD$

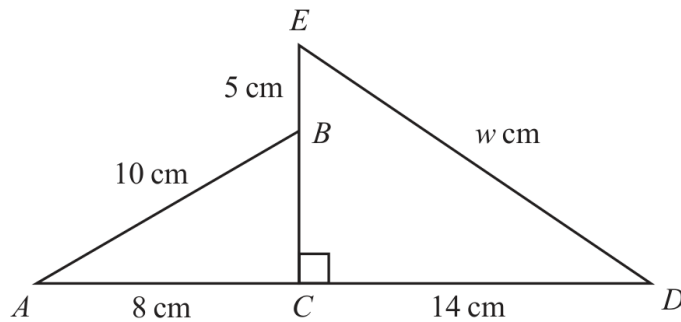


Diagram **NOT**  
accurately drawn

$ACD$  and  $EBC$  are straight lines.

$$AB = 10 \text{ cm} \quad AC = 8 \text{ cm} \quad EB = 5 \text{ cm} \quad CD = 14 \text{ cm} \quad ED = w \text{ cm}$$

Work out the value of  $w$

Give your answer correct to one decimal place.

$w = \dots\dots\dots$

**Worked solution 15**

Two right triangles joined at C

MODULAR UNIT 1

4 marks

**whole – Two right triangles joined at C**

Find BC

$$BC = \sqrt{10^2 - 8^2} = \sqrt{36} = 6.$$

Find EC

$$EC = EB + BC = 5 + 6 = 11.$$

Use the second right triangle

$$w^2 = ED^2 = EC^2 + CD^2 = 11^2 + 14^2 = 317.$$

Round the length

$$w = \sqrt{317} = 17.804\dots \approx 17.8 \text{ cm.}$$

**Final answer**

$$w = 17.8 \text{ cm}$$

**Question 16**

4MA1-2H Q4 (a)

ORIGINAL QUESTION UNIT 1

3 marks

4 (a) Solve  $\frac{2x+5}{6} = 2x-5$

Show clear algebraic working.

$x = \dots\dots\dots$   
(3)

**Worked solution 16**

Solve a linear equation with a fraction

MODULAR UNIT 1

3 marks

**(a) – Solve a linear equation with a fraction**

Clear the denominator

$$\frac{2x + 5}{6} = 2x - 5 \Rightarrow 2x + 5 = 6(2x - 5) = 12x - 30.$$

Collect terms

$$35 = 10x.$$

Solve

$$x = \frac{35}{10} = 3.5.$$

**Final answer**

$$x = 3.5$$

**Question 17**

4MA1-2H Q4 (b)

ORIGINAL QUESTION UNIT 1

1 marks

(b) Simplify  $h^{15} \div h^3$ .....  
(1)

**Worked solution 17**

Index law for division

MODULAR UNIT 1

1 marks

**(b) – Index law for division**

Subtract the powers

$$h^{15} \div h^3 = h^{15-3} = h^{12}.$$

**Final answer** $h^{12}$

**Question 18**

4MA1-2H Q4 (c)

ORIGINAL QUESTION UNIT 1

2 marks

(c) Simplify fully  $(2g^3k^5)^4$

.....  
(2)

**Worked solution 18**

Power of a product

MODULAR UNIT 1

2 marks

**(c) – Power of a product**

Raise each factor to the fourth power

$$(2g^3k^5)^4 = 2^4g^{3 \cdot 4}k^{5 \cdot 4}.$$

Simplify

$$(2g^3k^5)^4 = 16g^{12}k^{20}.$$

**Final answer** $16g^{12}k^{20}$



**Question 19**

4MA1-2H Q4 (d)

ORIGINAL QUESTION UNIT 1

2 marks

- (d) Given that  $\frac{y^5 \times y^n}{y^7} = y^{12}$   
work out the value of  $n$

$n = \dots\dots\dots$   
(2)

**Worked solution 19**

Find an unknown index

MODULAR UNIT 1

2 marks

**(d) – Find an unknown index**

Combine powers

$$\frac{y^5 y^n}{y^7} = y^{5+n-7} = y^{n-2}.$$

Equate exponents

$$n - 2 = 12 \Rightarrow n = 14.$$

**Final answer**

$$n = 14$$

**Question 20**

4MA1-2H Q6 whole

ORIGINAL QUESTION UNIT 1

3 marks

6 Show that  $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

**Worked solution 20**

Show a mixed-number division identity

MODULAR UNIT 1

3 marks

**whole – Show a mixed-number division identity**

Convert to improper fractions

$$3\frac{3}{7} = \frac{24}{7}, \quad 2\frac{2}{3} = \frac{8}{3}.$$

Invert and multiply

$$\frac{24}{7} \div \frac{8}{3} = \frac{24}{7} \times \frac{3}{8} = \frac{72}{56} = \frac{9}{7}.$$

Convert back to a mixed number

$$\frac{9}{7} = 1\frac{2}{7}.$$

**Final answer**

1 2/7

**Question 21**

4MA1-2H Q9 whole

ORIGINAL QUESTION UNIT 1

2 marks

- 9 A straight line passes through the points with coordinates  $(0, -3)$  and  $(2, 0)$

Find an equation of the line.

.....

**Worked solution 21**

Equation of a straight line

MODULAR UNIT 1

2 marks

**whole – Equation of a straight line**

Find the gradient

$$m = \frac{0 - (-3)}{2 - 0} = \frac{3}{2} = 1.5.$$

Use the y-intercept

The point  $(0, -3)$  gives  $c = -3$ , so  $y = 1.5x - 3$ .**Final answer**

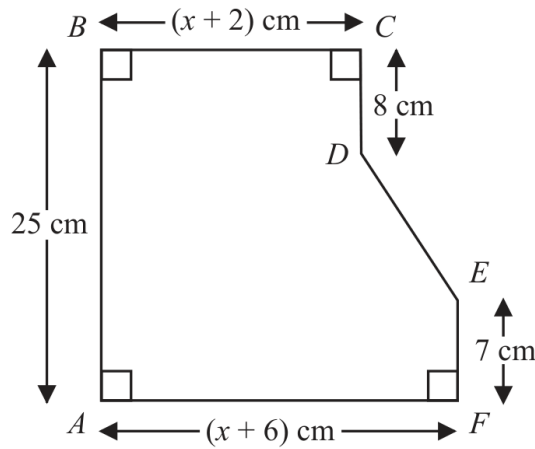
$$y = 1.5x - 3$$

**Question 22**

4MA1-2H Q10 whole

ORIGINAL QUESTION UNIT 1

5 marks

10 The diagram shows a hexagon  $ABCDEF$ Diagram **NOT**  
accurately drawn

$$AB = 25 \text{ cm} \quad BC = (x + 2) \text{ cm} \quad CD = 8 \text{ cm} \quad EF = 7 \text{ cm} \quad AF = (x + 6) \text{ cm}$$

The area of hexagon  $ABCDEF$  is  $258 \text{ cm}^2$ Work out the value of  $x$  $x = \dots\dots\dots$

**Worked solution 22**

Area of a stepped hexagon

MODULAR UNIT 1

5 marks

**whole – Area of a stepped hexagon**

Read the missing vertical lengths

$DE$  drops  $25 - 8 - 7 = 10$  cm; the horizontal width of the sloping strip is  $(x + 6) - (x + 2) = 4$  cm.

**Split into a rectangle and trapezium**

$$A = 25(x + 2) + \frac{1}{2}(17 + 7)(4) = 25(x + 2) + 48.$$

**Use the given area**

$$25(x + 2) + 48 = 258 \Rightarrow 25x + 98 = 258.$$

**Solve**

$$25x = 160 \Rightarrow x = 6.4.$$

**Final answer**

$$x = 6.4$$



**Question 23**

4MA1-2H Q11 (a), (b)

ORIGINAL QUESTION UNIT 1

5 marks

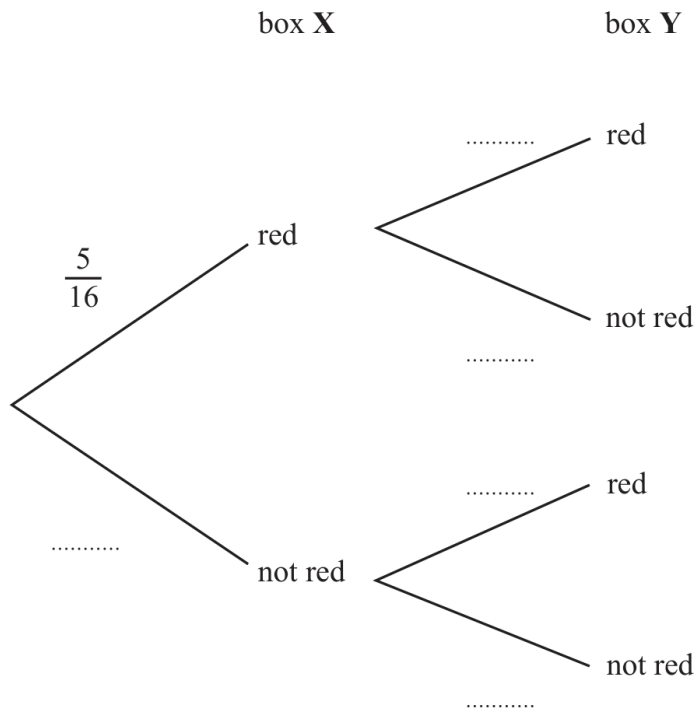
11 Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.

7 of the 20 crayons in box **Y** are red.

Sid takes at random one crayon from box **X** and one crayon from box **Y**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

(3)

**Worked solution 23**

Complete a two-box probability tree / Probability of two matching colours

**MODULAR UNIT 1****5 marks****(a) – Complete a two-box probability tree**

Complete box X

$$P(X \text{ not red}) = 1 - \frac{5}{16} = \frac{11}{16}.$$

Complete box Y on both branches

$$P(Y \text{ red}) = \frac{7}{20}, \quad P(Y \text{ not red}) = \frac{13}{20}.$$

**Final answer**

X not red = 11/16; each Y branch = 7/20 and 13/20

**(b) – Probability of two matching colours**

Add the two matching paths

$$P = \frac{5}{16} \cdot \frac{7}{20} + \frac{11}{16} \cdot \frac{13}{20}.$$

Combine

$$P = \frac{35 + 143}{320} = \frac{178}{320} = \frac{89}{160}.$$

**Final answer**

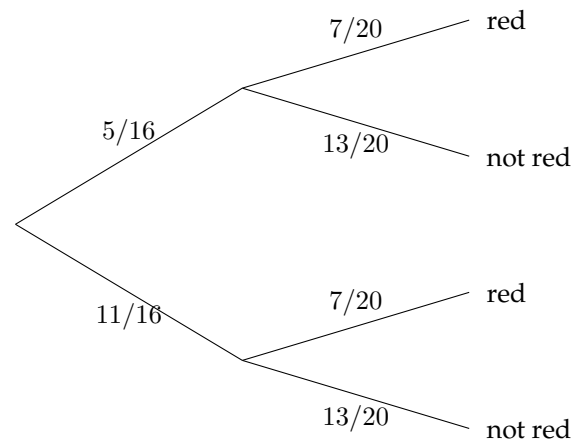
89/160

**Completed diagram 23**

Complete a two-box probability tree / Probability of two matching colours

MODULAR UNIT 1

5 marks

**Completed solution diagram: Completed probability tree**

**Question 24**

4MA1-2H Q12 (a)

ORIGINAL QUESTION UNIT 1

2 marks

**12**  $2^7 \times 4^5 = 4^x$

(a) Calculate the value of  $x$ 

$$x = \dots\dots\dots$$

(2)

**Worked solution 24**

Solve an index equation

MODULAR UNIT 1

2 marks

**(a) – Solve an index equation**

Write all terms in base 2

$$2^7 \cdot 4^5 = 2^7 \cdot (2^2)^5 = 2^{17}, \quad 4^x = (2^2)^x = 2^{2x}.$$

Equate exponents

$$2x = 17 \Rightarrow x = 8.5.$$

**Final answer**

$$x = 8.5$$

**Question 25**

4MA1-2H Q12 (b)

ORIGINAL QUESTION UNIT 1

2 marks

(b) Simplify fully  $(125p^6y^{24})^{\frac{2}{3}}$

.....  
(2)

**Worked solution 25**

Fractional power of a product

MODULAR UNIT 1

2 marks

**(b) – Fractional power of a product**

Apply the two-thirds power

$$(125p^6y^{24})^{2/3} = 125^{2/3}p^4y^{16}.$$

Simplify the coefficient

$$125^{2/3} = (5^3)^{2/3} = 5^2 = 25, \text{ hence } 25p^4y^{16}.$$

**Final answer**

$$25p^4y^{16}$$

**Question 26**

4MA1-2H Q15 (a)

ORIGINAL QUESTION UNIT 1

2 marks

15 (a) Use algebra to show that  $0.3\dot{7}\dot{2} = \frac{41}{110}$

(2)



**Worked solution 26**

Convert a recurring decimal to a fraction

MODULAR UNIT 1

2 marks

**(a) – Convert a recurring decimal to a fraction**

Set up the shift

$$x = 0.372727\dots, \quad 100x = 37.272727\dots$$

Subtract and simplify

$$99x = 36.9 = \frac{369}{10} \Rightarrow x = \frac{369}{990} = \frac{41}{110}.$$

**Final answer**

41/110

**Question 27**

4MA1-2H Q15 (b)

ORIGINAL QUESTION UNIT 1

3 marks

- (b) Express  $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$  in the form  $\sqrt{n}$  where  $n$  is an integer.

Show your working clearly.

.....  
(3)

**Worked solution 27**

Simplify a surd expression

MODULAR UNIT 1

3 marks

**(b) – Simplify a surd expression**

Simplify the numerator surds

$$\sqrt{125} + \sqrt{80} = 5\sqrt{5} + 4\sqrt{5} = 9\sqrt{5}.$$

Rationalise the denominator

$$\frac{9\sqrt{5}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{15}}{3} = 3\sqrt{15}.$$

Put it in the requested form

$$3\sqrt{15} = \sqrt{9 \cdot 15} = \sqrt{135}.$$

**Final answer** $\sqrt{135}$

**Question 28**

4MA1-2H Q16 whole

ORIGINAL QUESTION UNIT 1

3 marks

16 Expand and simplify  $(2x + 3)(x - 5)(x + 4)$

.....

**Worked solution 28**

Expand three brackets

MODULAR UNIT 1

3 marks

**whole – Expand three brackets**

Multiply two brackets

$$(x - 5)(x + 4) = x^2 - x - 20.$$

Multiply by the third bracket

$$(2x + 3)(x^2 - x - 20) = 2x^3 - 2x^2 - 40x + 3x^2 - 3x - 60.$$

Collect like terms

$$2x^3 + x^2 - 43x - 60.$$

**Final answer**

$$2x^3 + x^2 - 43x - 60$$

**Question 29**

4MA1-2H Q17 whole

ORIGINAL QUESTION UNIT 1

3 marks

**17**  $P = a(c + y)$

 $a = 8.3$  correct to 2 significant figures $c = 2$  correct to 1 significant figure $y = 15$  correct to the nearest 5Work out the upper bound for the value of  $P$ 

Show your working clearly.

.....

**Worked solution 29**

Upper bound of a product with a sum

MODULAR UNIT 1

3 marks

**whole – Upper bound of a product with a sum**

Choose upper bounds

8.3 to 2 s.f. gives  $8.25 \leq a < 8.35$ ; 2 to 1 s.f. gives  $1.5 \leq c < 2.5$ ; 15 to nearest 5 gives  $12.5 \leq y < 17.5$ .

Use the largest permitted values

$$P_{UB} = 8.35(2.5 + 17.5).$$

Evaluate

$$P_{UB} = 8.35(20) = 167.$$

**Final answer**

167

**Question 30**

4MA1-2H Q20 whole

ORIGINAL QUESTION UNIT 1

4 marks

- 20 The diagram shows equilateral triangle  $ABC$  with sides of length 10 cm. A circle is drawn inside the triangle.

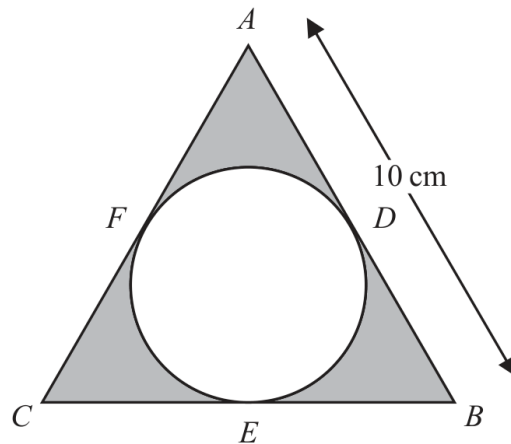


Diagram **NOT**  
accurately drawn

$D$ ,  $E$  and  $F$  are points on the circle.

$ADB$ ,  $BEC$  and  $CFA$  are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

.....  $\text{cm}^2$



**Worked solution 30**

Shaded area between an equilateral triangle and its incircle

MODULAR UNIT 1

4 marks

**whole – Shaded area between an equilateral triangle and its incircle**

Find the triangle area

$$A_{\triangle} = \frac{1}{2}(10)(10) \sin 60^{\circ} = 25\sqrt{3}.$$

Find the inradius

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}.$$

Subtract the circle area

$$A_{\text{shaded}} = 25\sqrt{3} - \pi r^2 = 25\sqrt{3} - \frac{25\pi}{3}.$$

Round

$$A_{\text{shaded}} = 17.121 \dots \approx 17.1 \text{ cm}^2.$$

**Final answer**17.1 cm<sup>2</sup>

**Question 31**

4MA1-2H Q24 whole

ORIGINAL QUESTION UNIT 1

5 marks

24 Solve  $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left( \frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$  .....

**Worked solution 31**

Solve a rational algebraic equation

MODULAR UNIT 1

5 marks

**whole – Solve a rational algebraic equation**

Combine the bracketed fractions

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x+3x-4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}.$$

Factor the first fraction

$$45x^3 - 80x = 5x(3x-4)(3x+4), \quad 3x^2 + x - 4 = (3x+4)(x-1).$$

Cancel common factors

The left-hand side simplifies to 20 (with the original denominator restrictions retained).

Solve the remaining equation

$$20 = \frac{4(x+2)}{5x-8} \Rightarrow 20(5x-8) = 4(x+2).$$

Find and check x

$$96x = 168 \Rightarrow x = \frac{7}{4}, \text{ which is not an excluded denominator value.}$$

**Final answer**

$$x = 7/4$$